

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
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Deep Learning

Section Id :	640653148263
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653417147
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532266893 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING (COMPUTER

BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406537369962. ✓ Yes

6406537369963. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653417148

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532266894 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

A neuron takes an input vector $x = [3.0, -2.0]^T$, weight vector $W = [0.5, 1.5]^T$, and bias $b = 0.5$. The neuron uses Parametric ReLU activation function parameterized by $\alpha = 0.1$:

$$f(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}.$$

Calculate the exact activation output $f(z)$ for this neuron. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-0.1

Question Number : 3 Question Id : 6406532266901 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

A satellite image of size $300 \times 300 \times 6$ is given as input to a convolutional layer. Suppose the layer uses 10 kernels, each of size $15 \times 15 \times 6$, and produces an output of size $96 \times 96 \times 10$. Assume zero padding (

$P = 0$). What is the value of the stride S ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Sub-Section Number : 3

Sub-Section Id : 640653417149

Question Shuffling Allowed : No

Question Id : 6406532266895 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (4 to 7)

Question Label : Comprehension

Suppose we want to minimize the following function:

$$f(x, y) = \frac{1}{2}(25x^2 + y^2), \quad x, y \in \mathbb{R}.$$

Let the initial point $(x_0, y_0) = (1, 1)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 4 Question Id : 6406532266896 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Using vanilla gradient descent with $\eta = 0.1$, compute the iterate (x_1, y_1) after one update, and enter your answer as $x_1 + y_1$. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-0.62 to -0.58

Question Number : 5 Question Id : 6406532266897 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Along which coordinate does gradient descent with $\eta = 0.1$ fail to converge?

Options :

6406537369966. ✖ y -coordinate

6406537369967. ✔ x -coordinate

6406537369968. ✖ Both coordinates diverge equally.

6406537369969. ✖ Neither; both converge at the same rate.

Question Number : 6 Question Id : 6406532266898 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

To adapt the learning rate separately for each coordinate, AdaGrad maintains

$$v_x^{(t)} = v_x^{(t-1)} + \left(\frac{\partial f}{\partial x}(x_t, y_t) \right)^2,$$

$$v_y^{(t)} = v_y^{(t-1)} + \left(\frac{\partial f}{\partial y}(x_t, y_t) \right)^2,$$

with $v_x^{(0)} = v_y^{(0)} = 0$. The effective learning rates are

$$\eta_x^{(t)} = \frac{\eta}{\sqrt{v_x^{(t)} + \epsilon}}, \quad \eta_y^{(t)} = \frac{\eta}{\sqrt{v_y^{(t)} + \epsilon}}.$$

For $\eta = 0.1$ and $\epsilon = 10^{-8}$, compute $\eta_x^{(1)} + \eta_y^{(1)}$.

Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.101 to 0.107

Question Number : 7 Question Id : 6406532266899 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

How does AdaGrad address the behavior observed with vanilla gradient descent?

Options :

6406537369971. ✓ The learning rate is reduced 25 times from the initial learning rate 0.1 along the x -direction.

6406537369972. ✗ The learning rate is reduced 25 times from the initial learning rate 0.1 along the y -direction.

6406537369973. ✗ The learning rate is reduced 5 times from the initial learning rate 0.1 along the y -direction.

6406537369974. ✗ The learning rate is reduced ⁵⁰ times from the initial learning rate ^{0.1} along both the x and y directions.

Sub-Section Number :

4

Sub-Section Id :

640653417150

Question Shuffling Allowed :

Yes

Question Number : 8 Question Id : 6406532266900 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a feedforward neural network with $L - 1$ hidden layers. The pre-activation at layer i is $a_i = W_i h_{i-1} + b_i$ and the activation is $h_i = g(a_i)$, where g is the activation function. Suppose every hidden-layer activation is the identity function $g(z) = z$, and the output activation is linear. Which of the following statements are correct?

Options :

6406537369975. ✓ The entire network computes a function of the form $\hat{y} = W'x + b'$ for some single matrix W' and vector b' , regardless of the number of hidden layers.

6406537369976. ✗ Adding more hidden layers with identity activations increases the representational capacity beyond that of a single-layer linear model.

6406537369977. ✓ Replacing the identity activation with a nonlinear function such as sigmoid enables the network to learn nonlinear decision boundaries.

6406537369978. ✗ With identity activations, the network can still learn arbitrary decision

boundaries provided enough hidden neurons are used.

Sub-Section Number : 5
Sub-Section Id : 640653417151
Question Shuffling Allowed : Yes

Question Number : 9 Question Id : 6406532266902 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

A multi-head attention block uses $d_{\text{model}} = 512$, $h = 8$ heads, and $d_k = 64$. Counting only the weight matrices W_Q^i, W_K^i, W_V^i (each of dimension $d_{\text{model}} \times d_k$) for every head, plus the output projection W_O (dimension $d_{\text{model}} \times d_{\text{model}}$), and ignoring all biases, compute the total number of parameters in this multi-head attention block.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1048576

Sub-Section Number : 6
Sub-Section Id : 640653417152
Question Shuffling Allowed : No

Question Id : 6406532266903 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 12)

Question Label : Comprehension

In a self-attention mechanism, each input vector has dimension D . The queries, keys, and values are computed using separate linear transformations, each mapping from $\mathbb{R}^D \rightarrow \mathbb{R}^D$. Suppose there are N input vectors.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 10 Question Id : 6406532266904 Question Type : MCQ Correct Marks : 3

Question Label : Multiple Choice Question

How many learnable parameters (including both weights and biases) are used to compute the

queries, keys, and values?

Options :

6406537369981. ✖ $3D^2$

6406537369982. ✔ $3D^2 + 3D$

6406537369983. ✖ $3D^2N + 3DN$

6406537369984. ✖ $3N^2 + 3N$

Question Number : 11 Question Id : 6406532266905 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many attention weights $a[i, j]$ are computed in total if we have $N = 100$ and $D = 16$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10000

Question Number : 12 Question Id : 6406532266906 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

How many learnable parameters (including both weights and biases) are there in a fully connected shallow network that maps all DN inputs to all DN outputs?

Options :

6406537369986. ✖ D^2N^2

6406537369987. ✔ $D^2N^2 + DN$

6406537369988. ✖ $(DN)^2 + D^2$

6406537369989. ✖ $(DN)^2 + D$

Sub-Section Number :

7

Sub-Section Id :

640653417153

Question Shuffling Allowed :

No

Question Id : 6406532266907 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (13 to 15)

Question Label : Comprehension

A sequence of length $T = 3$ has token embeddings arranged as rows of

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{T \times d_{\text{model}}}, \quad d_{\text{model}} = 3.$$

A single-head self-attention module computes queries, keys, and values as

$$Q = X W_Q, \quad K = X W_K, \quad V = X W_V,$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ with $d_k = 2$, so that $Q, K, V \in \mathbb{R}^{T \times d_k}$.

The i -th row of Q (respectively K, V) is the query q_i (respectively key k_i , value v_i) for token i . The parameter matrices are

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Scaled dot-product attention computes the output for query q_i as

$$z_i = \sum_{j=1}^T \alpha_{ij} v_j, \quad \alpha_{ij} = \frac{\exp(q_i \cdot k_j / \sqrt{d_k})}{\sum_{m=1}^T \exp(q_i \cdot k_m / \sqrt{d_k})},$$

with scale factor $\sqrt{d_k} = \sqrt{2}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 13 Question Id : 6406532266908 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

Which token receives the lowest dot-product score $q_1 \cdot k_j$ from query q_1 ? Enter the token index (1,

2, or 3).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 14 Question Id : 6406532266909 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Compute the attention weight α_{13} (the weight that query q_1 places on token 3). Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.483 to 0.489

Question Number : 15 Question Id : 6406532266910 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Compute the attention output vector

$z_1 = \begin{bmatrix} z_{11} \\ z_{12} \end{bmatrix}$ for query q_1 , and enter the sum of its two components $z_{11} + z_{12}$, correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.95 to 4.00

Sub-Section Number :

Sub-Section Id :

640653417154

Question Shuffling Allowed :

Yes

Question Number : 16 Question Id : 6406532266911 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a vanilla (non-attention) encoder-decoder model, the encoder compresses the entire input

sequence into a single fixed-dimensional vector s_0 , such as $s_0 = h_T$, where h_T is the final hidden state of an RNN encoder. The decoder uses this same vector to generate every output token. In practice, the translation quality of such a model tends to degrade as the length of the source sentence increases. Which of the following best explains this degradation? Select all that apply

Options :

6406537369993. ✖ The decoder's softmax layer has a fixed vocabulary size and therefore cannot generate longer sentences.

6406537369994. ✔ The fixed-dimensional vector s_0 becomes an information bottleneck as the source sentence length increases.

6406537369995. ✖ Teacher forcing is valid only for source sentences shorter than the maximum training length.

6406537369996. ✖ Longer source sentences necessarily require a higher-dimensional one-hot representation, which increases the input noise.

Sub-Section Number :

9

Sub-Section Id :

640653417155

Question Shuffling Allowed :

No

Question Id : 6406532266912 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (17 to 18)

Question Label : Comprehension

Consider a Skip-gram model with negative sampling. For a positive word-context pair (w, c) , suppose the loss for one training example is

$$L = -\log \sigma(w^T c) - \log \sigma(-w_{\text{neg}}^T c),$$

where σ is a logistic function. Suppose the initial vectors are

$$w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad w_{\text{neg}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The word vector w is updated using the gradient descent algorithm using a learning rate of $\eta = 0.5$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 17 Question Id : 6406532266913 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Calculate the loss L for the given positive word-context pair (w, c) and negative sample. Give your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.623 to 1.630

Question Number : 18 Question Id : 6406532266914 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

The word vector w is updated using the gradient descent algorithm. Using gradient descent, obtain the updated word vector w_{new} , and calculate the cosine similarity between w_{new} and c . Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.780 to 0.790

Sub-Section Number :

10

Sub-Section Id :

640653417156

Question Shuffling Allowed :

No

Question Id : 6406532266915 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (19 to 20)

Question Label : Comprehension

Consider a corpus containing $N = 10,000$ total words. Based on this, answer the given

subquestions:

Sub questions

Question Number : 19 Question Id : 6406532266916 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

A word w occurs $\text{count}(w) = 50$ times, a context c occurs $\text{count}(c) = 40$ times, and the pair (w, c) co-occurs $\text{count}(w, c) = 10$ times. Compute $\text{PMI}(w, c)$. Use natural log and enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.89 to 3.93

Question Number : 20 Question Id : 6406532266917 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

A second pair (w', c') has
 $\text{count}(w', c') = 2$, $\text{count}(w') = 200$,
 $\text{count}(c') = 300$.

Suppose its PMI is -1.0986 . Under the PPMI (positive PMI) transformation, what value is stored for this pair?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Sub-Section Number :

11

Sub-Section Id :

640653417157

Question Shuffling Allowed :

Yes

Question Number : 21 Question Id : 6406532266918 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A neural network is trained using mini-batch gradient descent. Suppose the mini-batch size is increased from 32 to 128 , while the learning rate and training dataset remain unchanged. Which of the following statements is correct?

Options :

6406537370001. ✖ Each update uses a gradient estimate with higher variance, while the number of updates per epoch decreases.

6406537370002. ✔ Each update uses a gradient estimate with lower variance, while the number of updates per epoch decreases.

6406537370003. ✖ The number of parameter updates per epoch increases by a factor of four, leading to faster convergence.

6406537370004. ✖ The larger batch size guarantees convergence to a better local minimum.